Change Detection in SAR Images Based on Progressive Nonlocal Theory

Huifu Zhuang, *Member, IEEE*, Hongdong Fan, Kazhong Deng, Kefei Zhang, Xuesong Wang, *Member, IEEE*, and Mengmeng Wang

***Abstract*— For multitemporal synthetic aperture radar (SAR) images, the change detection methods based on nonlocal theory can well suppress the adverse effects of coherent speckle noise over the change detection results. However, effectively retaining the edge information of the changed area is still a challenging task. To overcome this problem, this study proposes a change detection method based on progressive nonlocal theory. First, the progressive nonlocal theory is used to extract the spatial–temporal nonlocal information from multitemporal SAR images. Compared with the traditional nonlocal theory, the progressive nonlocal theory proposed in this study has three distinctive characteristics: 1) the progressive nonlocal neighborhood from the matching window to the search window; 2) the progressive optimization of matching window weight from the isotropic Gaussian distribution to the irregular distribution; and 3) the progressive increase of noise level from the 2 sigma principle to the 4/3 sigma principle (the noise level corresponding to the 4/3 sigma principle is 1.5 times the noise level corresponding to the 2 sigma principle). The difference image is then obtained by using the spatial–temporal nonlocal information and the ratio operator. Finally, the change map is obtained by applying a threshold segmentation method to the difference image. Two data sets were used for the testing, and it was shown that compared with other advanced methods, the method proposed in this study can better retain the edge information of the changed area and improve the Kappa coefficient and F1 score of the change map.**

***Index Terms*— Change detection, edge information, multitemporal, progressive nonlocal, synthetic aperture radar (SAR).**

I. Introduction

CHANGE detection exploits multitemporal remote sensing images covering the same geographic area to extract land cover change information [1]. It is one of the main research topics in earth observation and has attracted significant research interest in recent decades [2], [3]. This is partially due to its wide applications in many areas such as map updating [4], agricultural monitoring [5], forest monitoring [6], flood disaster assessment [7], and urban expansion research [8].

Compared with multispectral or hyperspectral satellites, synthetic aperture radar (SAR) satellites have the ability to acquire ground image all-time and all-weather. At present, SAR images have been widely used in earthquake disaster monitoring [9], urban subsidence analysis [10], environmental monitoring in mining areas [11], [12], agriculture mapping [13], glacier movement monitoring [14], etc. With the increasing demand for land cover change monitoring and the launch of more SAR satellites, change detection has gradually become one of the most important research directions in extracting information from SAR images [15]–[17].

For multitemporal remote sensing images, the change detection process is usually divided into three steps: 1) image preprocessing; 2) characterizing the difference of multitemporal images; and 3) obtaining a change map through difference analysis. Among these steps, characterization of the difference of multitemporal images is a key task. According to the image information used in the change detection of SAR images, the methods can be divided into two categories: 1) the change detection methods using the pixel-based information and 2) the change detection methods using the spatial neighborhood information. They are briefly introduced as follows.

1) Early research on change detection of SAR images mainly used pixel-based information. If the SAR image is not affected by the coherent speckle noise, the subtract method generally performs well in the change detection. Since the multiplicative noise model is good at processing the coherent information of SAR image [18], the ratio method was widely adopted in change detection [19], [20]. The log ratio (LR) method, developed on the basis of the ratio method, was a typical change detection method using the pixel-based information [21]–[23]. It uses logarithmic transformation to convert the multiplicative speckle noise of the SAR image into additive noise, which also makes the numerical range of the difference image well compressed. It is worth noting that the logarithmic transformation does not change the receiver operating characteristic (ROC) curve of the difference image. That is, the difference images obtained by the ratio and the LR methods have the same ROC curve [24]. In addition, Baronti *et al.* [25] studied a change detection method based on principal component analysis (PCA). After PCA transformation is performed

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on multitemporal SAR images, the first principal component contains the unchanged information of SAR images while the second principal component contains the changed information of SAR images. For the change detection of SAR images, the abovementioned methods have high computational efficiency, but are susceptible to coherent speckle noise.

2) With the development of computer hardware and image processing techniques, the SAR change detection methods using spatial neighborhood information have gradually attracted scholars' attention. In recent years, thanks to the development of deep learning technology, deep neural networks and convolutional neural networks (CNNs) have been researched and applied in the change detection of SAR images [15], [26]. However, the interpretability of the change detection method based on deep learning is poor. If the change detection method based on deep learning is trained on one data set, its change detection performance will usually decrease when it is applied to other data sets. This limits the applications of the change detection method based on deep learning. In addition, inspired by the PCA transformation theory and the change detection method using pixel-based information, Qiu *et al.* [27] studied a method based on neighborhood PCA for change detection in SAR images [27], whose performance is better than the pixel-based PCA method. Given the fact that the texture features of SAR image can reduce the impact of coherent speckle noise, the texture features based on gray-level co-occurrence matrix [28] and Gabor wavelet [29] were used in the study of change detection in SAR images. To use the spatial–temporal neighborhood information of multitemporal SAR images, Bujor *et al.* [30] studied a change detection method using the second-order logarithmic cumulant (SOLC). However, the SOLC will detect all the edge areas of the SAR image as changed areas. The mean ratio (MR) method proposed on the basis of the ratio method [31], and the LR using neighborhood information [23] is not sensitive to coherent speckle noise, but they perform poorly in areas with high heterogeneity such as edges. This is because MR and LR using neighborhood information do not consider the similarity of pixels in the local neighborhood. In order to overcome this problem, Gong *et al.* [32] and Zhuang *et al.* [33] studied the neighborhood-based ratio method, which uses heterogeneity to control the influence of neighborhood information on the center pixel. In addition, Pham *et al.* [34] reported a graph theory-based weighted log MR method, where the keypoints in a large neighborhood are used to avoid using a dense neighborhood. Different from the classic local mean, the extraction of nonlocal mean (NLM) is not limited to a single local neighborhood [35]. To extract the spatial neighborhood information of SAR image more reasonably and effectively, a ratio-based nonlocal information RNLI) was proposed and applied to change detection of SAR images [17]. RNLI has a good performance in change detection of SAR images, which can well suppress the influence of coherent speckle noise. In summary, the change detection methods using spatial neighborhood information of SAR images can reduce the influence of coherent speckle noise over the change detection result. However, for such methods, how to effectively retain the edge information of the changed area is still a challenging problem.

To overcome the problem mentioned above, this study proposes a method based on the progressive nonlocal theory for change detection in multitemporal SAR images. The general scheme of the proposed approach is shown in Fig. 1. First, the proposed progressive nonlocal theory is used to extract the progressive nonlocal information (PNLI) from the spatial–temporal neighborhood of the multitemporal SAR images. Different from the traditional NLM, PNLI uses a ratio-based Gaussian kernel function to characterize the similarity of two pixels in a nonlocal neighborhood. The noise level and matching window parameters used to extract PNLI will be automatically obtained, which is similar to the RNLI. Compared with RNLI, the extraction process of PNLI has the following three characteristics: 1) using the progressively increasing noise levels, which will be estimated based on the 2 sigma principle and the 4/3 sigma principle, respectively (the noise level corresponding to the 4/3 sigma principle is 1.5 times the noise level corresponding to the 2 sigma principle); 2) the matching window weight used to extract pixel similarity will be progressively optimized, whose distribution will be optimized from an isotropic Gaussian distribution to an irregular distribution (i.e., the stage 1 shown in Fig. 1); and 3) the size of the nonlocal neighborhood will be adjusted progressively. The nonlocal neighborhood with the same size as the matching window is used to optimize the weight of the matching window. PNLI is extracted from the nonlocal neighborhood with the size as the search window. Then, the difference image is generated by using the PNLI and the ratio operator. Finally, to get the change map, the threshold obtained by the manual trial-and-error (MTEP) method [17], [36] is used to segment the difference image, which is useful for objectively comparing different methods. The proposed method can extract spatial–temporal neighborhood information more reasonably and effectively, which has a significant advantage in retaining the edge information of changed areas.

This article is organized into five sections. The proposed method is shown in Section II. Experimental data sets and results are shown in Section III. Section IV presents the discussion of the experimental results and the proposed method. Finally, conclusions are drawn in Section V.

II. Methodology

A. RNLI

NLM method was designed to extract information from a nonlocal neighborhood of an image, which could suppress the impact of additive noise [35]. Since NLM has a good performance in extracting information, it has been applied to digital ground model generation [37], target tracking [38], medical image processing [39], [40], and video compression [41], [42], etc. However, the coherent speckle noise of an SAR image is usually modeled by a multiplicative noise model. In order to overcome this problem, Zhuang *et al.* [17] proposed

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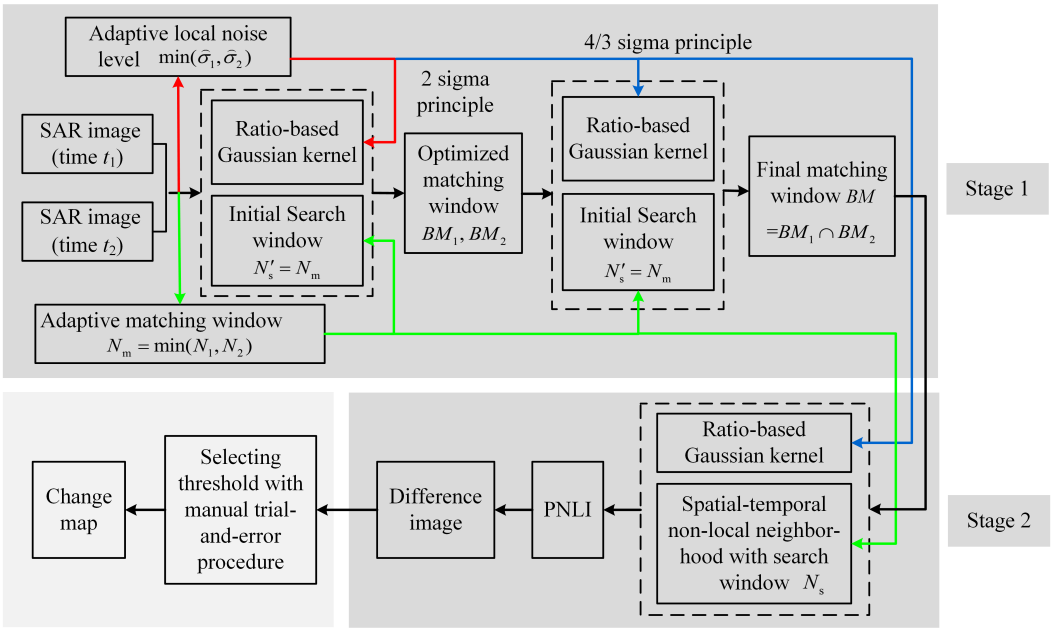


Fig. 1. General scheme of the proposed approach.

a RNLI method by using a ratio-based Gaussian kernel. For an image, the following equation is used to compute the RNLI at the position x:

$$u_{\mathrm{RNLI}}(x) = \sum_{i \in \Omega_x} w(i,x) I(i) \tag{1}$$

where $u_{\mathrm{RNLI}}(x)$ is the value of RNLI at the position x. $w(i,x)$ is the similarity weight of the pixels at positions i and x. $I(i)$ is the intensity value of the pixel at position i. Ω_x is a neighborhood composed by the pixels in the search window at the position x.

Unlike the traditional NLM, RNLI uses a ratio-based Gaussian kernel function to characterize the similarity of two pixels. This kernel function has two variables y_1 and y_2, which can be expressed by

$$G(y_1, y_2) = \exp\left(-\frac{(1 - \min(y_1, y_2)/\max(y_1, y_2))^2}{2\sigma^2}\right) \tag{2}$$

where σ represents the standard deviation of the ratio-based Gaussian kernel. Let $R(y_1, y_2) = \min(y_1, y_2)/\max(y_1, y_2)$, the curve of $G(y_1, y_2)$ changes with $R(y_1, y_2)$ is shown in Fig. 2. When σ takes different values, the greater the $R(y_1, y_2)$, the greater the $G(y_1, y_2)$, and the greater similarity between y_1 and y_2.

By using the ratio-based Gaussian kernel function, the similarity weight $w(i,x)$ can be obtained by

$$w(i,x) = \frac{1}{Z(x)} \exp\left(-\frac{||1 - \mathbf{R}||_{2,a}^2}{2\sigma^2}\right) \tag{3}$$

where $Z(x) = \sum_{i \in \Omega_x} \exp(-(||1 - \mathbf{R}||_{2,a}^2/2\sigma^2))$ is the normalization parameter. $a > 0$ is the standard deviation of a Gaussian kernel. $\mathbf{R} = \min(\mathbf{V}(i), \mathbf{V}(x))/\max(\mathbf{V}(i), \mathbf{V}(x))$ is the ratio vector between $\mathbf{V}(i)$ and $\mathbf{V}(x)$. $\mathbf{V}(i)$ and $\mathbf{V}(x)$ are the vectors at positions i and x, respectively, which are composed of pixels in the neighborhood with the same size as the matching window.

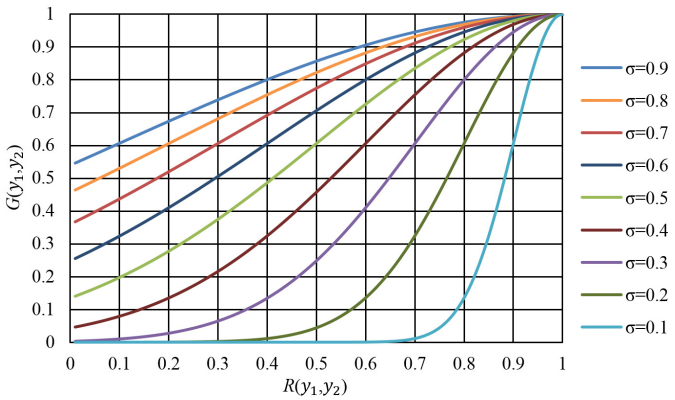


Fig. 2. Curve of $G(y_1, y_2)$ versus $R(y_1, y_2)$.

B. PNLI

The reliability of the extracted nonlocal information is mainly affected by the reliability of the similarity weight in the search window. The calculation of the similarity weight is related to the noise level and the homogeneity of the matching window. In this study, a PNLI method is proposed to improve the reliability of change detection in SAR images. After determining the sizes of the matching and search windows, the change detection method based on PNLI mainly includes two stages: 1) after progressive optimization, the first stage obtains the irregularly distributed weights of the matching window (i.e., the optimal homogeneity area in the matching window is selected and is used to calculate the similarity between the pixel in the search window and the center pixel) and 2) the second stage extracts the reliable nonlocal information and obtains the difference image based on PNLI.

The main steps of the first stage are as follows: first, set the size of the initial search window N_S' to be the same as the size of the matching window N_m [such as Fig. 3(a) and (f)]; second, use the isotropic Gaussian model with standard deviation a to determine the initial weight of the matching window [such as Fig. 3(b) and (g)], which is used to calculate the similarity

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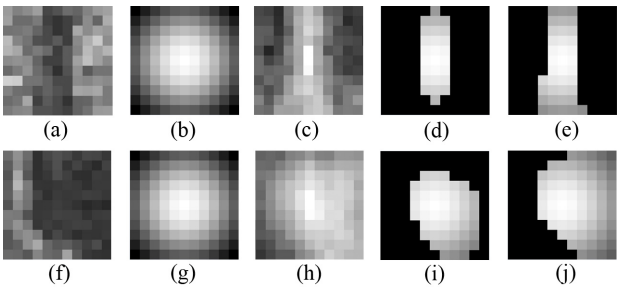


Fig. 3. Examples showing the progressive optimization of matching window weight: (a) and (f) matching windows with heterogeneity of 0.44 and 0.41, respectively; (b) and (g) initial weight of the matching window; (c) and (h) ratio image corresponding to $||\mathbf{R}||^2_{2,a}$ in (4); (d) and (i) weight image after the first optimization with the 2 sigma principle; and (e) and (j) weight image after the second optimization with the 4/3 sigma principle.

weight between the center pixel and other pixels in the search window; third, estimate the noise level σ according to the 2 sigma principle (the noise level estimation method will be introduced in Section II-C); and fourth, use the following equation to calculate the approximate similarity weight image AW_t of the pixels in the initial search window (with the same size as the matching window):

$$\mathrm{AW}_t(i,x) = \frac{1}{Z'_t(x)} \exp\left(-\frac{||1-\mathbf{R}||^2_{2,a}}{2\sigma^2}\right), \quad t = 1,2 \tag{4}$$

where $\mathrm{AW}_t(i,x)$ represents the similarity weight between the pixel at position i and the pixel at position x in the image I_t. $Z'_t(x) = \sum_{i\in\Omega'_x} \exp(-(||1-\mathbf{R}||^2_{2,a}/2\sigma^2))$, $t = 1,2$ is the normalization parameter, where Ω'_x represents the neighborhood composed of the pixels in the initial search window corresponding to the pixel at position x in the image I_t. The ratio image corresponding to $||\mathbf{R}||^2_{2,a}$ is shown in Fig. 3(c) and (h).

The similarity weight in AW_t is not accurate, which is affected by the noise level and the homogeneity of the matching window. On the basis of AW_t, this study will obtain the irregularly distributed weights in the homogeneity area of the matching window, which is related to the second characteristic of PNLI. The process of optimizing the matching window weight is as follows: first, a mean filter with a size of 3×3 is applied to AW_t; second, obtain a binary mask image BM_t of the homogeneity area with greater similarity by segmenting the filtered similarity weight image FAW_t with a threshold; third, after the initial optimization, the irregularly distributed weight image w_t [such as Fig. 3(d) and (i)] corresponding to the homogeneity area of the matching window is obtained by

$$w_t(i) = \frac{1}{Z'} \mathrm{BM}_t(i) \cdot G(i), \quad t = 1,2 \tag{5}$$

where $w_t(i)$ represents the weight of the pixel at the position i in the matching window corresponding to image I_t, which is used to calculate the similarity of the pixels in the search window. $Z' = \sum_{i\in\Omega'_x} \mathrm{BM}_t(i) \cdot G(i)$ is the normalization parameter. $G(i)$ is the weight of the pixel at the position i in the matching window and is also used to calculate similarity in (4), which is obtained by using an isotropic Gaussian model.

Then, repeat the steps of the first stage above [not including the use of (5)] to obtain a new binary mask image BM_t of the homogeneity area with greater similarity. The sections that different from the above steps are as follows: use the 4/3 sigma principle instead of the 2 sigma principle to get the estimated noise level σ; for (4), w_t is used to replace the initial isotropic weight of the matching window. For the multitemporal SAR images, let $\mathrm{BM} = \mathrm{BM}_1 \cap \mathrm{BM}_2$, we can get the mask image BM corresponding to the homogeneity area in the multitemporal SAR images. Finally, the irregularly distributed weight image w [such as Fig. 3(e) and (j)], in the homogeneity area of the matching window, is obtained by

$$w(i) = \frac{1}{Z} \mathrm{BM}(i) \cdot G(i) \tag{6}$$

where $w(i)$ represents the weight of the pixel at the position i in the matching window, which is used to calculate the similarity of the pixels in the search window. $Z = \sum_{i\in\Omega'_x} \mathrm{BM}(i) \cdot G(i)$ is the normalization parameter.

Next, we will introduce the second stage of PNLI, which will use the noise level σ estimated according to the 4/3 sigma principle. The main steps of the second stage are as follows: first, the difference image based on PNLI is obtained by the following equation, of which the optimized irregularly distributed matching window weight image w and the determined search window N_S will be used:

$$D_{\mathrm{PNLI}}(x) = 1 - \frac{\min\{u_{\mathrm{PNLI1}}(x), u_{\mathrm{PNLI2}}(x)\}}{\max\{u_{\mathrm{PNLI1}}(x), u_{\mathrm{PNLI2}}(x)\}} \tag{7}$$

where D_{PNLI} represents the difference image obtained by using PNLI; $D_{\mathrm{PNLI}}(x)$ represents the value of the pixel at the position x in D_{PNLI}; $u_{\mathrm{PNLI1}}(x)$ represents the PNLI at the position x in image I_1; and $u_{\mathrm{PNLI2}}(x)$ represents the PNLI at the position x in image I_2. In D_{PNLI}, the unchanged pixel has a small intensity value while the changed pixel has a large intensity value.

In this study, $u_{\mathrm{PNLI1}}(x)$ is calculated by the following equation [similarly, we can get $u_{\mathrm{PNLI2}}(x)$]:

$$u_{\mathrm{PNLI1}}(x) = \sum_{i\in\Omega''_x} w''(t,i,1,x) I(t,i), \quad t = 1,2 \tag{8}$$

where $I(t,i)$ represents the intensity value of the pixel at position i in image I_t; Ω''_x represents the spatial–temporal neighborhood composed of the search windows corresponding to the pixels at the position x in the multitemporal images I_1 and I_2; and $w''(t,i,1,x)$ is the weight that characterizes the similarity between the pixel at the position i in image I_t and the pixel at the position x in image I_1. $w''(t,i,1,x)$ can be obtained by the following equation, of which the ratio-based Gaussian kernel function and the matching window weight $w(i)$ obtained by progressive optimization will be used:

$$w''(t,i,1,x) = \frac{1}{Z''(1,x)} \exp\left(-\frac{||\mathbf{w}\cdot(1-\mathbf{R})||^2_2}{2\sigma^2}\right), \quad t = 1,2 \tag{9}$$

where $Z''(1,x) = \sum_{i\in\Omega''_x} \exp(-(||\mathbf{w}\cdot(1-\mathbf{R})||^2_2/2\sigma^2))$, $t = 1,2$ is the normalization parameter. $\mathbf{w}$ is a vector composed of the matching window weights $w(i)$, and the value of $\mathbf{w}$ at position i is equal to $\sqrt{w(i)}$, which makes the expansion of $||\mathbf{w}\cdot(1-\mathbf{R})||^2_2$ is equal to $\sum w_i(1-\mathbf{R}(i))^2$. $\mathbf{R} = \min(\mathbf{V}(t,i), \mathbf{V}(1,x))/\max(\mathbf{V}(t,i), \mathbf{V}(1,x))$ is the ratio vector

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between $\mathbf{V}(t,i)$ and $\mathbf{V}(1,x)$. $\mathbf{V}(t,i)$ and $\mathbf{V}(1,x)$ represent the vectors containing the pixels in a neighborhood (equal in size to the matching window) at the positions i (image I_t) and x (image I_1), respectively.

In summary, the PNLI-based change detection method can be described by Algorithm 1.

Algorithm 1 Change Detection Based on PNLI

Begin
(a) Input the multi-temporal SAR images I_1 and I_2.
(b) Set sigma principle couple: 2 sigma and 4/3 sigma; set the minimum and maximum values of the matching window: $N_{\min}$ and $N_{\max}$; set the search window N_s.
(c) For each pixel of the SAR image, adaptively select the matching window size N_m with the method used in the literature [43];
(d) For the pixel at position x in the image I_t, set the initial search window $N_s' = N_m$.
(e) Set the window size of mean filter as 3×3.
(f) Initially optimize the matching window based on N_s' and 2 sigma principle, obtain the irregularly distributed weight image w_t.
(g) Perform final optimization of the matching window based on w_t, N_s' and 4/3 sigma principle, get the final matching window weight image w.
(h) Compute and save the difference value $D_{\mathrm{PNLI}}(x)$ based on w, N_s, and 4/3 sigma principle until the complete difference image D_{PNLI}. has been generated.
(i) Output the change map by applying a threshold method to D_{PNLI}.
End

C. Parameters Setting of PNLI

The noise level σ is one of the main parameters of the PNLI-based change detection method. In this study, the σ of ratio-based Gaussian kernel will be adaptively estimated using the method proposed in [17]. The 1 sigma principle, the 2 sigma principle, and the 3 sigma principle are important features of the Gaussian model. Assuming that the mean and standard deviation of a Gaussian random variable g are u_G and σ_G, respectively. The 1 sigma principle refers to that the probability of $|g-u_G| < \sigma_G$ is 68.27%, the 2 sigma principle refers to that the probability of $|g-u_G| < 2\sigma_G$ is 95.45%, and the 3 sigma principle refers to that the probability of $|g-u_G| < 3\sigma_G$ is 99.73%. Therefore, Zhuang *et al.* [17] suggest using $\hat{\sigma}_R(x)/2$ or $\hat{\sigma}_R(x)/3$ as the noise level parameter, which, respectively, correspond to the 2 sigma principle and 3 sigma principle in noise level estimation. In this study, the noise levels used in PNLI will progressively increase, which will be estimated based on the 2 sigma principle and the 4/3 sigma principle, respectively.

The sizes of the matching and search windows are important parameters in nonlocal theory. To further improve the degree of automation and reliability of extracting PNLI, this study will employ the method used in [43] to realize the adaptive selection of the matching window. This method is based on the heterogeneity ∂ of the neighborhood (the standard deviation of a neighborhood divided by its mean is equal to the value of ∂). Usually, a window with the ∂ smaller than a given threshold is regarded as an optimal matching window. Given the fact that using a large window to calculate the heterogeneity is more reliable than a small window [43], [44]. In this study, for a given value range $[N_{\min}, N_{\max}]$ of the optimal matching window, the window search strategy from large to small is used to determine the optimal matching window of the center pixel. The value range $N_{\min} = 5$, $N_{\max} = 11$ of matching window recommended by Zhuang *et al.* [17] will be used in this study. If the reader wants to know more details of the algorithm for adaptive selection of the optimal window, they can read [43]. In addition, this study will use the 21×21 search window recommended by Buades *et al.*[35], [45].

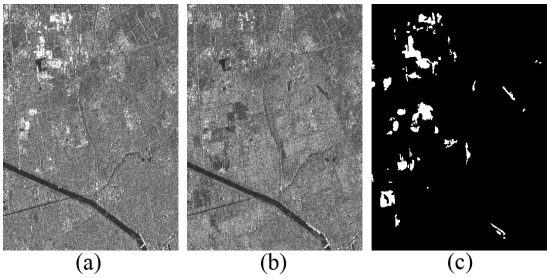


Fig. 4. Multitemporal images of data set 1: (a) image acquired on June 13, 2017; (b) image acquired on June 25, 2017; and (c) map of changed areas (ground truth) used as the reference in the experiments.

To reduce the uncertainty of the change detection result caused by the artificial threshold, the threshold used in the first stage of the PNLI method needs to be automatically selected. The number to select thresholds for PNLI is the same as the number of pixels contained in the SAR image. Therefore, it is recommended to use the threshold automatic selection methods with high efficiency, such as OTSU (named after the author's last name) [46], and the maximum entropy method [47]. In this study, the OTSU is used to automatically select the threshold. It is worth noting that in some cases, the initial search window is of high homogeneity, and the difference between the similarity weights in FAW_t is small. To avoid the strong segmentation of FAW_t, it is recommended to expand the edge of FAW_t with a size of 1 before segmentation. The extended edge needs to be assigned a value smaller than the minimum value $\min(\mathrm{FAW}_t)$ in FAW_t. Typically, the pixel value of the extended area can be made equal to $0.9 \times \min(\mathrm{FAW}_t)$. In addition, the extended edge will be reduced after segmentation.

III. Experimental Data Sets and Results

The experiments in this study were carried out on two data sets acquired by Terra-SAR-X and Sentinel-1 satellites. In order to effectively verify the effectiveness of the PNLI in change detection of SAR images, the pixel-based LR [22]–[24], SOLC [30], neighborhood-based LR [23], LR using nonlocal information (NL-LR) [35], [48], RNLI [17], and deep learning method based on CNN (DL-CNN) [15] were used as comparison methods.

A. Experimental Data

The experiments of this study were carried out on two data sets. The first data set (shown in Fig. 4) is the SAR images acquired by Sentinel-1 satellite in Peixian, China. The size of the image is 346×486, and the acquisition time was June 13 and 25, 2017, respectively. The second data set is shown in Fig. 5 (with a size of 800×700), which was obtained

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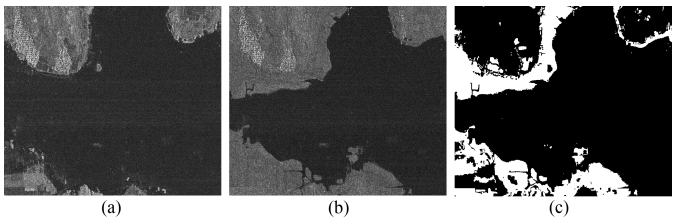


Fig. 5. Multitemporal images of data set 2: (a) image acquired on May 2015; (b) image acquired on November 2015; and (c) map of changed areas (ground truth) used as the reference in the experiments.

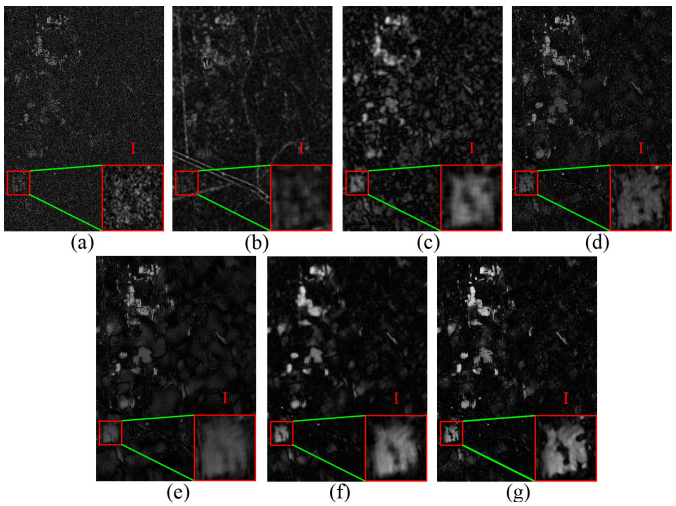


Fig. 6. Difference images obtained with: (a) LR; (b) SOLC 5 × 5; (c) LR 7 × 7; (d) NL-LR 5 × 5; (e) NL-LR 11 × 11; (f) RNLI; and (g) PNLI in the data set 1.

by Terra-SAR-X satellite in May and November 2015, and is located in Handan, China. Figs. 4(c) and 5(c) were acquired through an accurate and time-consuming visual inspection of the SAR images. These images are called reference change maps, which are used to quantitatively compare the change detection performance of different methods.

B. Difference Images of Data Set 1

The difference images obtained through different methods in data set 1 are shown in Fig. 6. In the initial experiments of this study, for SOLC and neighborhood-based LR methods, the windows from 3 × 3 to 11 × 11 were used to obtain the difference images. For data set 1, the experimental results show that the SOLC with 5 × 5 window and the LR with 7 × 7 window have good performance, and the corresponding results are shown in Fig. 6. RNLI and PNLI used the same matching window parameters: $N_{\min} = 5$ and $N_{\max} = 11$. To better prove the performance of the proposed PNLI method, NL-LR with matching windows of 5 × 5 and 11 × 11 was used as the comparison method. In addition, it can be seen that there is no difference image corresponding to DL-CNN in Fig. 6 because this method directly obtains the change map through the neural network.

The red rectangle I in Fig. 6 shows an enlarged view of a small area, in which an unchanged area is surrounded by changed pixels. In the difference image obtained by the LR [Fig. 6(a)], it is difficult to identify the changed and unchanged areas in the rectangle I. This is because LR does not use the spatial neighborhood information of the SAR image.

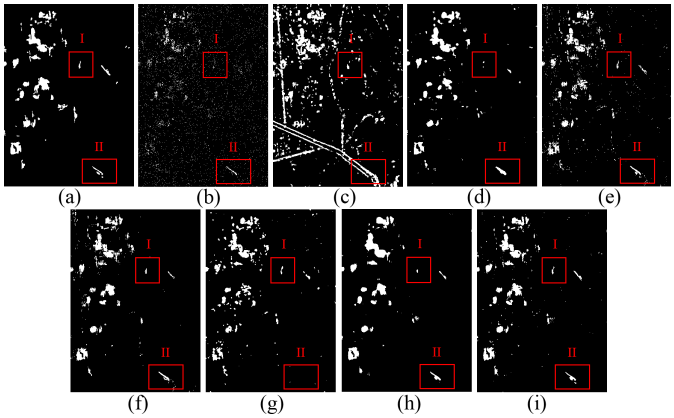


Fig. 7. Change maps obtained with: (a) ground truth; (b) LR; (c) SOLC 5 × 5; (d) LR 7 × 7; (e) NL-LR 5 × 5; (f) NL-LR 11 × 11; (g) DL-CNN; (h) RNLI; and (i) PNLI in the data set 1.

In the difference image obtained by the SOLC [Fig. 6(b)], the contrast between the changed and unchanged areas in the red rectangle I is low. Compared with the difference images obtained by the NL-LR [Fig. 6(d) and (e)], the unchanged area with fuzzy edges can be seen from the red rectangle I in Fig. 6(c) and (f). In Fig. 6(d), the edge of the changed area in the red rectangle I is clearer than that in Fig. 6(e), which shows that the NL-LR with a smaller matching window can better retain the edge information of the changed area. It can be seen from Fig. 6 that the changed and unchanged areas obtained by the PNLI [Fig. 6(g)] are easiest to distinguish through the clear edge contours. Compared with other methods, PNLI can effectively extract spatial–temporal information from multitemporal SAR images, which is benefit from the progressive nonlocal theory. For data set 1, the PNLI can suppress the influence of coherent speckle noise on change detection, and it has a good performance in preserving the edge information of the changed area.

C. Change Maps of Data Set 1

The change maps corresponding to different methods in data set 1 are shown in Fig. 7. The DL-CNN can directly obtain the change map, while other methods first obtain the difference images (as shown in Fig. 6) and then obtain the change maps through the reasonable segmentation threshold. Although there are some methods that can automatically calculate the threshold that meet certain assumptions (for example, the difference image obeys a Gaussian mixture distribution) [49]–[51], it is uncertain whether *a priori* assumption is applicable to a data set [33], [51]. In this study, to avoid the influence of the selection error of optimal threshold on the experimental analysis, the MTEP method widely used in the change detection literature [17], [33], [36], [52] was used to determine the optimal threshold.

In the change maps shown in Fig. 7, both red rectangles I and II contain a long and narrow changed area. In the change map obtained by the LR [Fig. 7(b)], the changed area in the red rectangle I is not well-detected, while a long and narrow changed area is detected in the red rectangle II. In addition, in the red rectangle I of Fig. 7(b), some discrete pixels in the unchanged area are incorrectly identified as changed pixels, which are affected by the coherent speckle noise of

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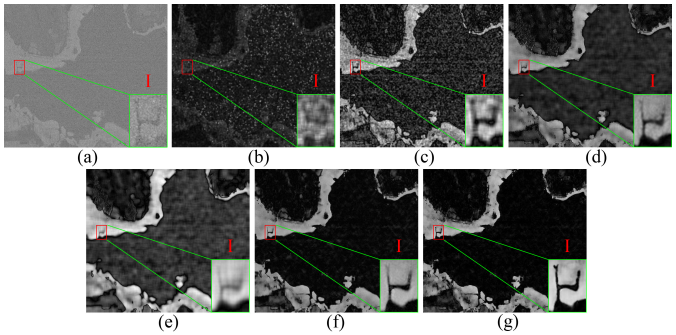


Fig. 8. Difference images obtained with: (a) LR; (b) SOLC 7 × 7; (c) LR 7 × 7; (d) NL-LR 5 × 5; (e) NL-LR 11 × 11; (f) RNLI; and (g) PNLI in the data set 2.

the SAR image. Fig. 7(c) shows the change map obtained by the SOLC; it can be seen that there are some edge areas which are identified as changed areas. This is because SOLC has a shortcoming that the pixels in heterogeneous areas will be recognized as changed pixels. For other change detection methods that use spatial neighborhood information of SAR image, in the red rectangle I, the number of changed pixels recognized by the LR 7 × 7, NL-LR 11 × 11, and RNLI is less than that of NL-LR 5 × 5, DL-CNN, and PNLI. Compared with NL-LR 5 × 5 and PNLI, DL-CNN failed to recognize the changed area contained in the red rectangle II. Compared with PNLI, NL-LR 5 × 5 recognizes a small number of unchanged pixels in the red rectangle I as changed pixels. The qualitative analysis of the experimental results shows that the change detection performance of PNLI on data set 1 is better than other comparison methods, which benefits from the progressive nonlocal theory proposed in this study.

D. Difference Images of Data Set 2

The difference images obtained through different methods in data set 2 are shown in Fig. 8. The neighborhood-based LR, NL-LR, RNLI, and PNLI methods used the same parameters as data set 1. Different from the data set 1, SOLC with 7 × 7 window is suitable for data set 2. In Fig. 8, the red rectangle I contains an inverted h-shaped unchanged area. It is a challenging problem to accurately detect this long and narrow unchanged area. Fig. 8(a) shows the difference image obtained by the LR; the contour of the unchanged area in the rectangle I could be roughly seen, but the contrast between the changed and unchanged areas is very low. Fig. 8(b) shows the difference image obtained by the SOLC; there are many unchanged areas with large intensity values, which are related to the characteristic of SOLC identifying the changed area and have been introduced in Section III-B. Fig. 8(c) shows the difference image obtained by neighborhood-based LR; the contrast of the unchanged area and the changed area in the rectangle I is greater than that in Fig. 8(a). This is because neighborhood-based LR uses the spatial neighborhood information of the SAR image. Fig. 8(d) and (e) shows the difference images obtained through NL-LR. For the difference image obtained by NL-LR 5 × 5, the shape of the unchanged area in the rectangle I is clearer than that in the difference image obtained by NL-LR 11 × 11, which verifies the prior knowledge that a small window can better retain the edge information of the changed area. For the change detection

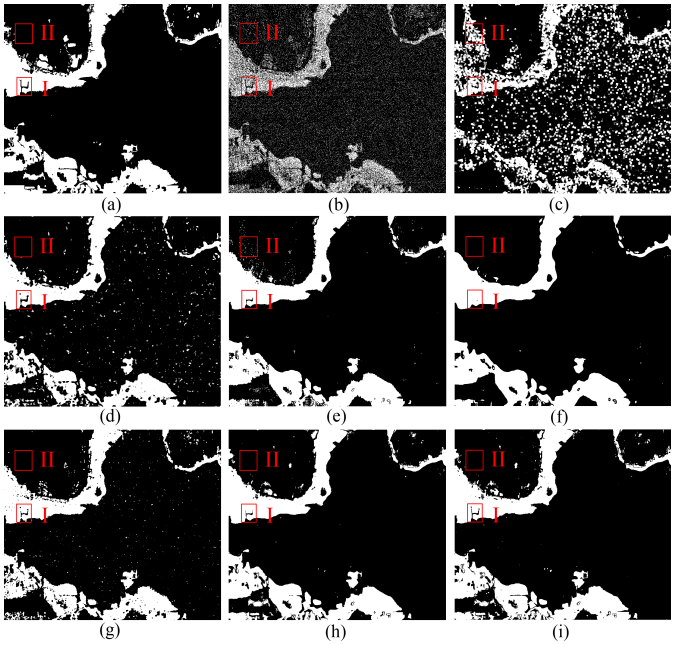


Fig. 9. Change maps obtained with: (a) ground truth; (b) LR; (c) SOLC 7 × 7; (d) LR 7 × 7; (e) NL-LR 5 × 5; (f) NL-LR 11 × 11; (g) DL-CNN; (h) RNLI; and (i) PNLI in the data set 2.

methods using spatial neighborhood information, the contours of the inverted h-shaped unchanged area in the red rectangle I corresponding to NL-LR 5 × 5, RNLI, and PNLI are clearer than those in SOLC, LR 7 × 7, and NL-LR 11 × 11. Comparing all the difference images in Fig. 8, the PNLI [Fig. 8(g)] generates the clearest contour of the inverted h-shaped unchanged area. In addition, the contrast between the changed area and the unchanged area in the red rectangle I corresponding to PNLI is greater than that of other methods. In summary, the experimental results show that the PNLI based on progressive nonlocal theory could effectively improve the accuracy of the edge information of the changed area.

E. Change Maps of Data Set 2

Fig. 9 shows the change maps obtained through different methods in data set 2. The threshold selection method used on data set 1 was also applied to data set 2. There are two red rectangles I and II in the difference images shown in Fig. 9. The red rectangle I contains an inverted h-shaped unchanged area, and the red rectangle II contains a large unchanged area. Fig. 9(b) shows the change map obtained through the LR; many unchanged pixels in the red rectangle II are incorrectly detected as changed pixels. This is mainly because LR does not use the spatial neighborhood information of the SAR image. For the change map obtained by SOLC [Fig. 9(c)], a large number of unchanged pixels in the red rectangle II are identified as changed pixels. This is because SOLC erroneously recognizes the pixels in the heterogeneous area as changed pixels. Compared with the change map obtained by LR, the pixels in the red rectangle II corresponding to the change map obtained by LR 7 × 7 are correctly detected as unchanged pixels. However, LR can detect the inverted h-shaped unchanged area in the red rectangle I, while LR 7 × 7 can only correctly detect a part of the unchanged area. Fig. 9(e) and (f) shows the change maps obtained by

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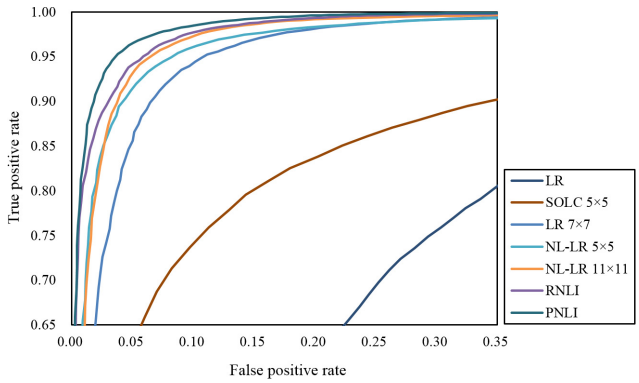


Fig. 10. ROC curves of the difference images corresponding to data set 1.

NL-LR with the matching windows of 5 × 5 and 11 × 11. The NL-LR 5 × 5 can correctly detect most of the unchanged pixels in the red rectangle I, while the NL-LR 11 × 11 can only correctly detect a small number of unchanged pixels. Comparing all the change maps in Fig. 9, it can be seen that PNLI can detect the unchanged area of the inverted h-shaped in the red rectangle I, so it has an obvious advantage in retaining the edge information of the changed area.

IV. Discussion

A. Discussion on the Results of Data Set 1

ROC curve [53], [54] was used to compare the difference images obtained by different methods in the research related to change detection. The characteristic of a good change detection method is that the ROC curve corresponding to this method is close to the top left corner [24]. The ROC curves obtained through different methods in data set 1 are shown in Fig. 10. As the ROC curve of LR is at the bottom right of the ROC curves of other methods, the LR method is worse than the other methods that use spatial neighborhood information. Although SOLC performs better than LR in the change detection of data set 1, the ROC curve of SOLC is located at the bottom right of the ROC curves of other methods. In addition, the ROC curve of the proposed PNLI is closest to the top left corner, which shows that PNLI has the best change detection performance in data set 1.

The Kappa–threshold curve (KTC) of a difference image reflects the relationship between the Kappa and the threshold [17]. On the one hand, the larger the maximum Kappa of a method, the better the change detection performance of this method. On the other hand, the smaller the curvature of a method's KTC near the maximum Kappa, the lower the sensitivity of its change detection performance to the threshold change. The KTCs of different methods in data set 1 are shown in Fig. 11. Similar to the results displayed by the ROC curve, the change detection performance of LR and SOLC is worse than that of other methods, and their maximum Kappa is smaller than that of other methods. The curvature of the KTCs of PNLI and RNLI near the maximum Kappa is smaller than that of other methods. Although the maximum Kappa of KTC corresponding to RNLI is larger than that of LR, SOLC, LR 7 × 7, and NL-LR, it is smaller than that corresponding to PNLI. This is because PNLI uses progressive nonlocal theory, which can better retain the edge information of the changed area.

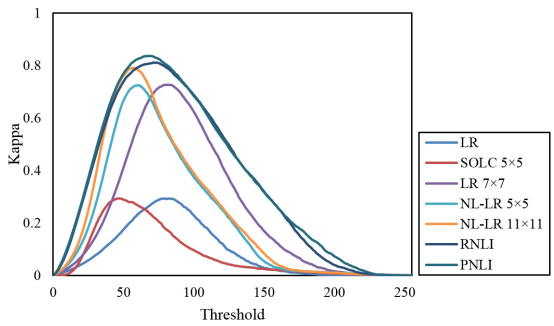


Fig. 11. KTCs of the difference images corresponding to data set 1.

TABLE I

False Alarms, Missed Alarms, Overall Error (in the Number of Pixels), Kappa, and F1 Score of the Change Maps Obtained With Different Methods in Data Set 1

Methods	False Alarms	Missed Alarms	Overall Error	Kappa	F1 Score
LR	3194	4837	8031	0.291	0.316
SOLC 5×5	12355	3023	15378	0.292	0.331
LR 7×7	1384	2001	3385	0.724	0.735
NL-LR 5×5	1465	1973	3475	0.722	0.733
NL-LR 11×11	1062	1604	2666	0.784	0.792
DL-CNN	1475	1945	3420	0.725	0.735
RNLI	797	1512	2309	0.811	0.818
PNLI	**738**	**1277**	**2015**	**0.837**	**0.843**

In this study, the quantitative analysis of the change map is based on false alarms, missed alarms, overall error, Kappa, and F1 score. A good change map has large Kappa and F1 score, small false alarms, missed alarms, and overall error. Table I shows the quantitative analysis indexes of the change maps obtained in data set 1. It can be seen that the false alarms of SOLC are larger than those of other methods because this method recognizes the edge heterogeneity area as a changed area. However, the missed alarms of SOLC are less than that of LR, and the Kappa is slightly larger than that of LR. This is because the neighborhood information of SAR image used by SOLC can reduce the impact of coherent speckle noise. There is significant difference between the Kappa of NL-LR 5 × 5 and the Kappa of NL-LR 11 × 11, which reflects that the change detection performance of NL-LR is affected by the size of matching window. The Kappa of DL-CNN is greater than that of LR 7 × 7 and SOLC, which indicates that the method based on deep learning technology has certain competitiveness in change detection of SAR images. In Table I, the best value of each indicator is shown in bold. It is easy to see that in data set 1, the change detection performance of the proposed PNLI is significantly better than that of other comparison methods. Compared with RNLI, the good change detection performance of PNLI is benefit from the use of progressive nonlocal theory.

B. Discussion on the Results of Data Set 2

For data set 2, the ROC curves corresponding to different methods are shown in Fig. 12. Similar to the results in data set 1, the ROC curves corresponding to LR and SOLC are at the bottom right of the ROC curves of other methods. However, the ROC curve of SOLC is closest to the bottom

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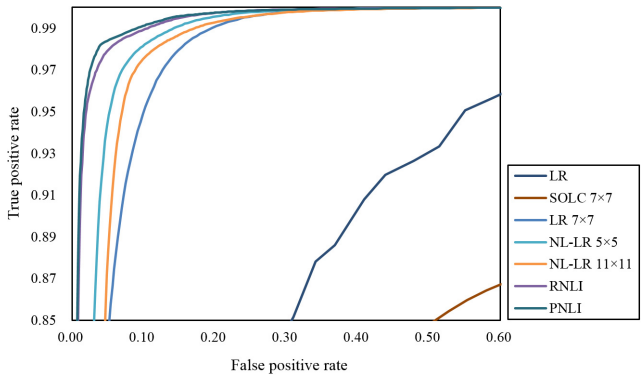


Fig. 12. ROC curves of the difference images corresponding to data set 2.

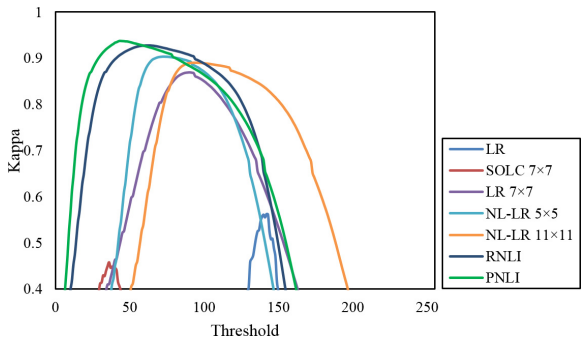


Fig. 13. KTCs of the difference images corresponding to data set 2.

right corner because many unchanged pixels in data set 2 are located in heterogeneity area. In addition, for data sets 1 and 2, the ROC curve of NL-LR with appropriate matching window is at top left of the ROC curves of LR 7 × 7 and SOLC, which shows that the nonlocal information has a good prospect in change detection of SAR images. However, different from the results in data set 1, the ROC curve of NL-LR 11 × 11 is at the bottom right of the ROC curve corresponding to NL-LR 5 × 5, which shows that the appropriate matching window size of NL-LR is not fixed on different data sets. Compared with RNLI, PNLI based on progressive nonlocal theory improves the reliability of the edge information of the changed area. Hence, the ROC curve performance of RNLI is worse than that of PNLI. Moreover, the ROC curve corresponding to PNLI is closest to the top left corner, which indicates that the change detection performance of PNLI on data set 2 is better than those of other methods.

The KTCs of different methods in data set 2 are shown in Fig. 13. Similar to the results reflected by the ROC curve, the performance of SOLC is worse than that of other methods because the maximum Kappa of SOLC is the smallest. The maximum Kappa of LR is significantly smaller than that of neighborhood-based LR, which is because LR 7 × 7 uses the spatial neighborhood information of SAR image. It can be seen that the maximum Kappa of RNLI is larger than that of NL-LR, which gets benefit from the use of the ratio-based Gaussian kernel. Since the PNLI uses the progressive nonlocal theory to improve the reliability of the edge information of the changed area, its maximum Kappa value is larger than that of RNLI. In addition, the curvature of PNLI's KTC near the maximum Kappa is small, which shows that it is easy to select a threshold to obtain a change map of the Kappa close to the maximum.

Table II shows the quantitative analysis indexes of the change maps obtained from data set 2. It can be seen that the change detection performance of LR is better than SOLC, which is consistent with the results reflected by ROC curve and KTC. The Kappa and F1 score of LR 7 × 7 are greater than those of LR, which is due to the fact that LR 7 × 7 utilizes the local neighborhood information of SAR images. Different from the results on data set 1, the change detection performance of NL-LR 5 × 5 is better than that of NL-LR 11 × 11, which shows that the appropriate matching window size of NL-LR is different on different data sets. The Kappa and F1 score of DL-CNN are greater than those of LR 7 × 7 and NL-LR 11 × 11, which shows that DL-CNN based on deep learning technology is a very competitive method for change detection of SAR images. If the five quantitative analysis indicators in Table II are comprehensively considered, the PNLI based on progressive nonlocal theory performs best in the change detection of data set 2. This is because the proposed PNLI can well retain the edge information of the changed area.

TABLE II

FALSE ALARMS, MISSED ALARMS, OVERALL ERROR (IN THE NUMBER OF PIXELS), KAPPA, AND F1 SCORE OF THE CHANGE MAPS OBTAINED WITH DIFFERENT METHODS IN DATA SET 2

Methods	False Alarms	Missed Alarms	Overall Error	Kappa	F1 Score
LR	41415	45303	86718	0.563	0.601
SOLC 7×7	79374	46842	126216	0.446	0.503
LR 7×7	8467	17074	25541	0.869	0.880
NL-LR 5×5	7662	11636	19298	0.902	0.911
NL-LR 11×11	7914	13879	21793	0.889	0.899
DL-CNN	7285	13635	20920	0.894	0.903
RNLI	7981	6574	14555	0.927	0.935
PNLI	**7203**	**5472**	**12675**	**0.937**	**0.943**

C. Discussion on the Effect of Sigma Couple

The matching window size and the noise level are the two key parameters of PNLI. In this study, the matching window size is adaptively selected by the method reported in [43]. Thus, the performance of PNLI is mainly affected by the noise level, which is determined based on the sigma principle couple (2 sigma and 4/3 sigma have been introduced in Section II-C). To present the effect of sigma principle couple on PNLI, the PNLI was applied to data set 2 under 13 different sigma couples. First, we set the first sigma parameter as a fixed 2 sigma when the second sigma parameter is set as 1/3, 2/3, 3/3, 4/3, 5/3, 6/3, 7/3, and 8/3 sigma, respectively. Second, we set the second sigma parameter as a fixed 4/3 sigma when the first sigma parameter is set as 0.5, 1, 1.5, 2, 2.5, and 3 sigma, respectively.

The maximum Kappa of PNLI under different sigma couples is shown in Fig. 14. When the first sigma parameter is fixed as 2 sigma, the maximum Kappa of PNLI increases first and then decreases with the increase of the second sigma parameter. Similarly, when the second sigma parameter is fixed as 4/3 sigma, the maximum Kappa of PNLI increases first

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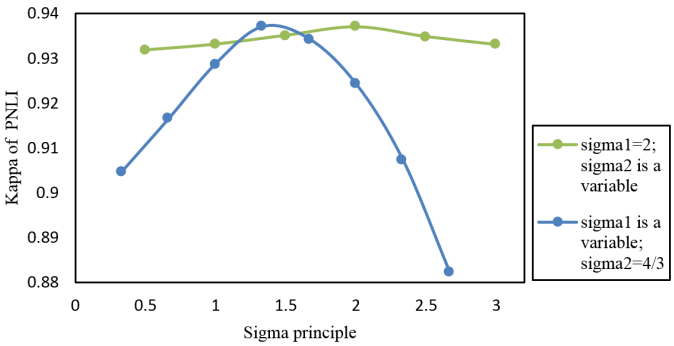


Fig. 14. Maximum Kappa of PNLI under different sigma couples.

and then decreases with the increase of the second sigma parameter. When using 2 sigma and 4/3 sigma couple, the maximum Kappa of PNLI is larger than that of other sigma couples. It is worth noting that when the first sigma parameter is fixed as 2 sigma, the change of PNLI's maximum Kappa is smaller than that when the second sigma parameter is fixed as 4/3 sigma. Therefore, compared with the first sigma parameter, the performance of PNLI is sensitive to the second sigma parameter. This is because the second sigma parameter directly affects the reliability of the nonlocal information extracted by PNLI.

D. Discussion on the Computational Complexity

The computational complexity of a method is a function of the problem size n [55], [56]. The discussion on the computational complexity of the proposed PNLI and other traditional methods is useful for readers to better understand the proposed method.

For the change detection problem in this study, n represents the number of pixels in SAR image. For each pixel in SAR image, the classic LR method includes three operations (i.e., ratio operation, logarithmic operation, and absolute value operation). If LR is applied to an SAR image with n pixels, the sum of the operations is $3n$. Thus, the computational complexity of LR is $O(n)$, which shows that the computational cost of LR is a linear function with respect to n. For each pixel in SAR image, the computation equations of SOLC, neighborhood-based LR, NL-LR, RNLI, and PNLI are independent of n and could be decomposed into several subalgorithms. For example, the neighborhood-based LR could be decomposed into four main subalgorithms: the mean of the local neighborhood, the ratio of multitemporal means, the logarithmic transformation, and the absolute value operation. Given the fact that the computational complexity of the four main subalgorithms of neighborhood-based LR is $O(n)$, the computational complexity of neighborhood-based LR is $O(n)$. This is because if an algorithm consists of several subalgorithms of computational complexity $O(n)$, then its computational complexity is also $O(n)$ [55]. Similarly, we could confirm the computational complexity of SOLC, NL-LR, RNLI, and PNLI is also $O(n)$. For each pixel in SAR image, the PNLI could be decomposed into eight main subalgorithms: adaptively select the matching window size, determine the initial weight of the matching window, adaptively estimate the noise level, calculate the similarity weight image with (4), a mean filter, obtain the irregularly distributed weight image by (5), get the optimized final matching window weight image by (6), and compute the difference value by (7)–(9). The computational complexity of the eight main subalgorithms is $O(n)$ because they are independent of n. Therefore, the computational complexity of the proposed PNLI is $O(n)$, which is the same as the comparison methods.

E. Discussion on the Proposed Method

For change detection in SAR images, the popular methods use the spatial neighborhood information to suppress the influence of coherent speckle noise, but it also causes an inaccurate edge information of the changed area. To solve this problem, this study proposes a PNLI method based on progressive nonlocal theory and compares it with other advanced methods. Experiments on two data sets prove the effectiveness of PNLI in change detection of SAR images. The search window, matching window, and noise level are the three important parameters of nonlocal theory. Compared with the traditional nonlocal theory, the "progressiveness" of the proposed method is mainly reflected in three aspects.

1) The progressive adjustment of the search window (i.e., the nonlocal neighborhood). In the edge heterogeneity area of an SAR image, the use of an isotropic matching window will reduce the reliability of the extracted nonlocal information. The coherent speckle noise contained in SAR images increases the difficulty of determining the parameters of the anisotropic Gaussian model. For this reason, this study uses nonlocal theory to improve the reliability of the extracted nonlocal information. Specifically, this study uses the nonlocal theory to select the optimal homogeneity area in the matching window, which could reduce the adverse effect of nonhomogeneous pixels in the matching window on the extraction of nonlocal information. Therefore, in the first stage of extracting PNLI, make the size of the search window the same as the size of the matching window. In the second stage of extracting PNLI, the real search window is used to extract reliable nonlocal information.
2) The progressive optimization of the matching window weight. When extracting nonlocal information, the similarity between the pixel in the search window and the center pixel is computed based on the matching window. The higher the similarity between the center pixel and other pixels in the matching window, the more reliable the similarity weight calculated based on the matching window. However, an automatically selected optimal matching window sometimes contains a small number of nonhomogeneous pixels. This is because there are some anisotropic neighborhoods in SAR images (for example, the neighborhoods of the pixels located at edge area). To solve this problem, this study proposes a progressive optimization method for matching window weights from isotropic Gaussian distribution to irregular distribution. Specifically, this method is based on nonlocal theory, takes the matching window as a nonlocal neighborhood, and uses progressively increasing noise levels to obtain

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the optimal homogeneity area within the matching window that do not use preset shape constraints and obtain the optimized window weights of irregular distribution.

3) The progressive increase of the noise level. The noise level parameter will affect the reliability of nonlocal information extraction, which is obtained by an automatic estimation method in this study. The proposed PNLI method uses the progressively increasing noise levels, which will be obtained based on the 2 sigma principle and the 4/3 sigma principle, respectively (the noise level corresponding to the 4/3 sigma principle is 1.5 times the noise level corresponding to the 2 sigma principle). Due to the low reliability of the initial weights of the matching window, PNLI first uses the noise level estimated based on the 2 sigma principle to obtain the initial optimal homogeneity area in the matching window. Then, the noise level estimated based on the 4/3 sigma principle is used to obtain the final optimal homogeneity area in the matching window. On this basis, the irregularly distributed optimized weights of the matching window are obtained. Finally, the difference image based on PNLI is obtained by using the noise level estimated based on the 4/3 sigma principle. It is worth noting that three or more noise levels can also be used within the framework of the PNLI method. Compared with the RNLI method, different noise levels are used for the progressive optimization of the matching window weight. However, the experiments we have performed show that three or more noise levels had little effect on the experimental results but increase the running time of the algorithm. The experiments also reflect that the reliability of the matching window weight has been greatly improved when two different noise levels are used. This is because PNLI uses a ratio-based Gaussian kernel, which could better characterize the similarity of SAR pixels with multiplicative speckle noise.

V. Conclusion

For change detection of multitemporal SAR images, this study proposes a PNLI method based on progressive nonlocal theory. Similar to the change detection methods based on nonlocal theory, PNLI has the advantage that the extraction of neighborhood information is not limited to a local neighborhood. Different from the existing change detection methods that use nonlocal information, PNLI uses nonlocal theory to solve the problem that a small number of heterogeneous pixels in the matching window affect the edge information retention of the changed area. Compared with the traditional nonlocal theory, the proposed progressive nonlocal theory has three distinctive characteristics (i.e., the progressive adjustment of the nonlocal neighborhood, the progressive optimization of the matching window weight, and the progressive increase of the noise level), which makes PNLI has a good ability to retain the edge information of the changed area.

In this study, the experiments were carried out on two data sets to prove the effectiveness of the proposed PNLI. The experimental results obtained by PNLI and other six change detection methods were compared qualitatively and quantitatively. The experimental results show that: 1) the change detection performance of PNLI is better than that of other comparison methods and 2) the advantage of PNLI's change detection performance is different on different data sets, the greater the proportion of changed pixels located at the edge of the changed area to all changed pixels, the greater the advantage of PNLI. This study contributes to: 1) propose a progressive nonlocal theory, which uses nonlocal theory to solve the problem that the nonlocal information extracted in edge heterogeneity area is of low reliability and 2) propose a PNLI-based change detection method for multitemporal SAR images, which can improve the accuracy of the edge information of the changed area.

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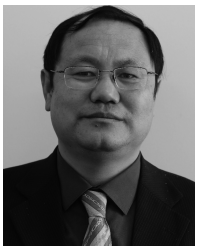
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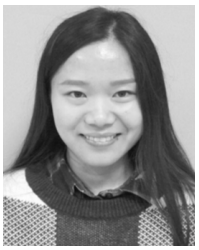
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